

Day 25 – Answer Intent and Understanding Engine

1. Objective

The Answer Intent and Understanding Engine processes candidate responses from the screening workflow and converts them into structured semantic information.

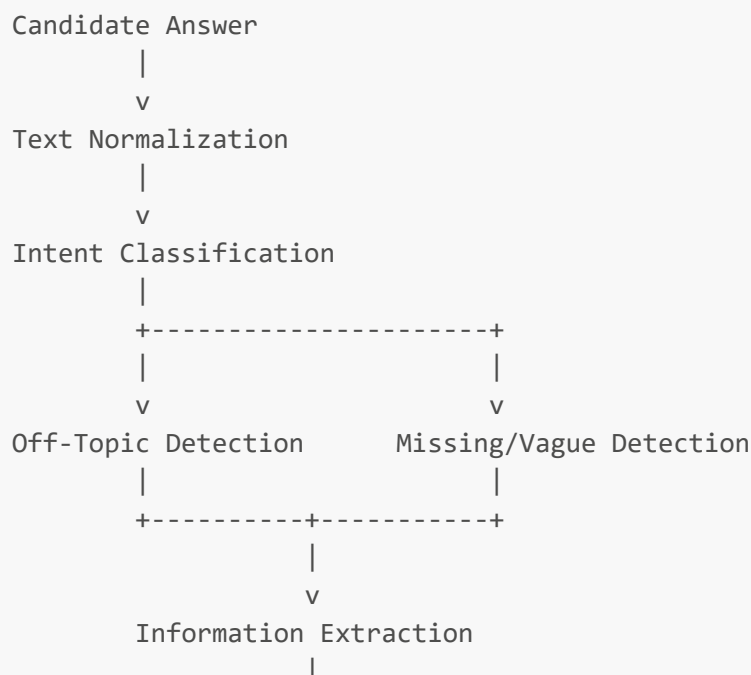
The engine identifies the likely intent of a candidate answer, detects vague or off-topic responses, and extracts basic information relevant to the identified intent.

2. Processing Capabilities

The Day 25 engine supports:

- Answer intent classification
- Answer text normalization
- Skills intent detection
- Experience intent detection
- Availability intent detection
- Salary expectation intent detection
- Off-topic response detection
- Missing or vague answer detection
- Skill extraction
- Experience information extraction
- Salary information extraction
- Structured semantic output

3. Answer Understanding Flow



V Structured Semantic Object

4. Text Normalization

The engine normalizes candidate answers before processing.

Normalization includes:

- Removing leading and trailing whitespace
- Converting multiple whitespace characters into single spaces
- Converting text to lowercase for matching

This provides a consistent representation for keyword-based intent detection.

5. Intent Classification

The engine classifies candidate responses into supported intent categories.

Skills

Used when the candidate discusses technical or professional skills.

Example:

"I have skills in Python, SQL and Power BI"

Expected intent:

`skills`

Experience

Used when the candidate discusses previous work or professional experience.

Example:

"I have three years of experience in data analysis"

Expected intent:

`experience`

Availability

Used when the candidate discusses joining or availability.

Example:

"I am available to join immediately"

Expected intent:

availability

Salary Expectation

Used when the candidate discusses expected compensation.

Example:

"My expected salary is 6 LPA"

Expected intent:

salary_expectation

6. Keyword-Based Intent Scoring

Intent classification uses configured keywords stored in:

`data/answer_intent_patterns.json`

The engine calculates a matching score for each configured intent.

The intent with the highest matching score is selected as the primary intent.

If no configured keywords match, the engine returns:

unknown

with a confidence value of `0.0`.

7. Confidence Calculation

The engine generates a confidence value based on the number of matching intent keywords.

The confidence is capped at `0.95`.

This provides a consistent confidence value for downstream processing.

8. Off-Topic Detection

The engine detects responses that contain configured off-topic keywords.

For example, a response such as:

"I watched a cricket match yesterday"

can be identified as an off-topic response.

The structured result marks:

- `off_topic` as `True`
- `response_status` as `off_topic`

9. Missing or Vague Answer Detection

The engine identifies empty, extremely short, or vague responses.

Examples include:

- Yes
- No
- Maybe
- Not sure
- I don't know
- None
- Nothing
- Some
- Okay
- Fine

Such responses are marked as missing or vague.

The structured result contains:

```
missing_or_vague: True
```

and:

```
response_status: missing_or_vague
```

10. Skill Extraction

For answers classified with the `skills` intent, the engine checks for known technical skills.

Supported examples include:

- Python
- SQL
- Java
- JavaScript
- TypeScript
- C++
- C#
- HTML
- CSS
- Excel
- Power BI
- Tableau
- Machine Learning
- Deep Learning
- Pandas
- NumPy
- Scikit-learn
- Docker
- Kubernetes

The extracted skills are stored in the semantic data section of the response.

11. Experience Extraction

For answers classified with the **experience** intent, the engine extracts numerical experience values expressed in years.

Example:

"I have 3 years of experience in data analysis"

produces:

years: [3]

The original normalized answer is also retained as experience details.

12. Salary Extraction

For answers classified with the **salary_expectation** intent, the engine extracts numerical salary values.

The implementation supports salary expressions involving:

- LPA
- Lakhs
- Lakh
- INR
- Rs.
- ₹

Example:

"I expect around 6 LPA"

produces a salary value containing:

6

13. Structured Semantic Object

The **understand()** method combines the processing stages into a structured result.

The result contains:

- Raw answer text
- Normalized answer text
- Detected intent
- Confidence
- Off-topic status
- Missing/vague status
- Response status
- Semantic data

This structure can be used by downstream AI screening components.

14. Answer Understanding Engine Alias

The implementation also provides:

`AnswerUnderstandingEngine`

as an alias derived from `AnswerIntentEngine`.

This allows the same implementation to be represented using answer-understanding terminology within the screening architecture.

15. Implementation

Main implementation:

`screening_ai/answer_intent_engine.py`

Intent configuration:

`data/answer_intent_patterns.json`

Automated tests:

`tests/test_answer_intent_engine.py`

16. JSON Encoding Handling

During testing, the intent-pattern JSON file contained a UTF-8 BOM.

The JSON loading implementation was updated to use:

`utf-8-sig`

This allows the engine to correctly read the JSON configuration while supporting UTF-8 BOM encoded files.

17. Testing

The Day 25 automated test suite validates:

- Pattern file availability
- Skills intent classification
- Experience intent classification
- Availability intent classification
- Salary expectation classification
- Skill extraction
- Experience extraction
- Salary extraction
- Off-topic detection
- Vague answer detection
- Structured semantic object generation

- Empty answer handling
- Multiple answer types

18. Test Result

All Day 25 Answer Intent Engine tests passed successfully.

13 passed in 0.07s

19. Day 25 Deliverable

The completed Answer Intent and Understanding Engine provides a structured interpretation layer for candidate screening responses.

It converts candidate answers into normalized, classified, and structured semantic information that can be used by downstream screening and analysis components.